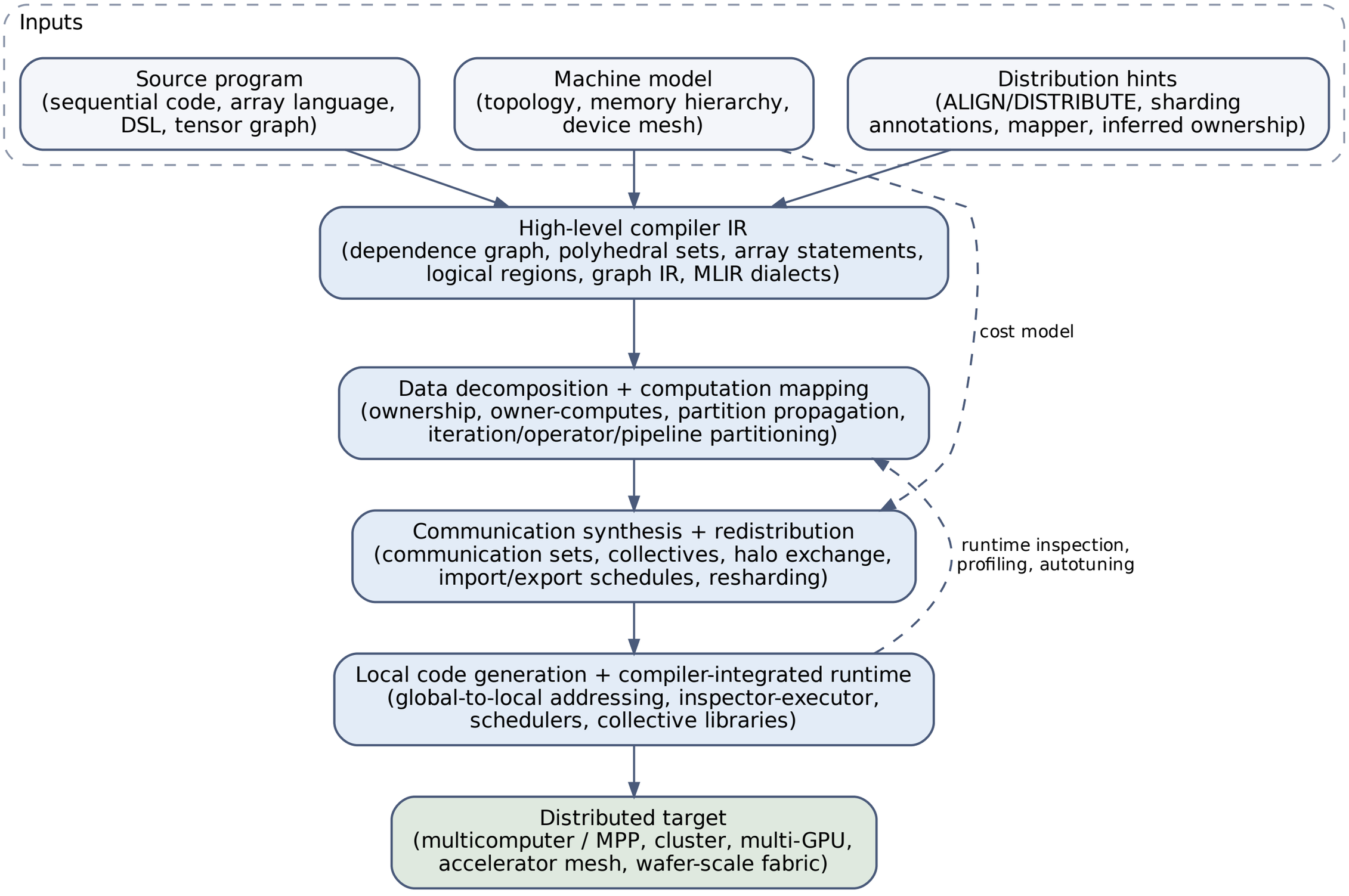
Inputs
Source program
(sequential code, array language,
DSL, tensor graph)
Machine model
(topology, memory hierarchy,
device mesh)
Distribution hints
(ALIGN/DISTRIBUTE, sharding
annotations, mapper, inferred ownership)
High-level compiler IR
(dependence graph, polyhedral sets, array statements,
logical regions, graph IR, MLIR dialects)
Data decomposition + computation mapping
(ownership, owner-computes, partition propagation,
iteration/operator/pipeline partitioning)
Communication synthesis + redistribution
(communication sets, collectives, halo exchange,
import/export schedules, resharding)
Local code generation + compiler-integrated runtime
(global-to-local addressing, inspector-executor,
schedulers, collective libraries)
Distributed target
(multicomputer / MPP, cluster, multi-GPU,
accelerator mesh, wafer-scale fabric)
cost model
runtime inspection,
profiling, autotuning